

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

**PHYSICAL SETTING
CHEMISTRY**

Tuesday, June 23, 2015 — 9:15 a.m. to 12:15 p.m., only

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

This is a test of your knowledge of chemistry. Use that knowledge to answer all questions in this examination. Some questions may require the use of the *2011 Edition Reference Tables for Physical Setting/Chemistry*. You are to answer *all* questions in all parts of this examination according to the directions provided in this examination booklet.

A separate answer sheet for Part A and Part B–1 has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet. Record your answers to the Part A and Part B–1 multiple-choice questions on this separate answer sheet. Record your answers for the questions in Part B–2 and Part C in your separate answer booklet. Be sure to fill in the heading on the front of your answer booklet.

All answers in your answer booklet should be written in pen, except for graphs and drawings, which should be done in pencil. You may use scrap paper to work out the answers to the questions, but be sure to record all your answers on your separate answer sheet or in your answer booklet as directed.

When you have completed the examination, you must sign the statement printed on your separate answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet and answer booklet cannot be accepted if you fail to sign this declaration.

Notice. . .

A four-function or scientific calculator and a copy of the *2011 Edition Reference Tables for Physical Setting/Chemistry* must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

Part A

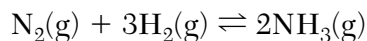
Answer all questions in this part.

Directions (1–30): For *each* statement or question, record on your separate answer sheet the *number* of the word or expression that, of those given, best completes the statement or answers the question. Some questions may require the use of the *2011 Edition Reference Tables for Physical Setting/Chemistry*.

- | | |
|---|---|
| <p>1 Compared to an electron, which particle has a charge that is equal in magnitude but opposite in sign?
(1) an alpha particle (3) a neutron
(2) a beta particle (4) a proton</p> <p>2 The mass of a proton is approximately equal to
(1) 1 atomic mass unit
(2) 12 atomic mass units
(3) the mass of 1 mole of carbon atoms
(4) the mass of 12 moles of electrons</p> <p>3 Which property <i>decreases</i> when the elements in Group 17 are considered in order of increasing atomic number?
(1) atomic mass (3) melting point
(2) atomic radius (4) electronegativity</p> <p>4 Any substance composed of two or more elements that are chemically combined in a fixed proportion is
(1) an isomer (3) a solution
(2) an isotope (4) a compound</p> <p>5 Which term refers to how strongly an atom of an element attracts electrons in a chemical bond with an atom of a different element?
(1) entropy
(2) electronegativity
(3) activation energy
(4) first ionization energy</p> <p>6 At STP, which substance has metallic bonding?
(1) ammonium chloride (3) iodine
(2) barium oxide (4) silver</p> | <p>7 What is the number of electrons shared between the carbon atoms in a molecule of ethyne?
(1) 6 (3) 8
(2) 2 (4) 4</p> <p>8 Which atom in the ground state has a stable valence electron configuration?
(1) Ar (3) Si
(2) Al (4) Na</p> <p>9 What occurs when two fluorine atoms react to produce a fluorine molecule?
(1) Energy is absorbed as a bond is broken.
(2) Energy is absorbed as a bond is formed.
(3) Energy is released as a bond is broken.
(4) Energy is released as a bond is formed.</p> <p>10 Which gas sample at STP has the same number of molecules as a 2.0-liter sample of $\text{Cl}_2(\text{g})$ at STP?
(1) 1.0 L of $\text{NH}_3(\text{g})$ (3) 3.0 L of $\text{CO}_2(\text{g})$
(2) 2.0 L of $\text{CH}_4(\text{g})$ (4) 4.0 L of $\text{NO}(\text{g})$</p> <p>11 All atoms of uranium have the same
(1) mass number
(2) atomic number
(3) number of neutrons plus protons
(4) number of neutrons plus electrons</p> <p>12 The concentration of a solution can be expressed in
(1) kelvins
(2) milliliters
(3) joules per kilogram
(4) moles per liter</p> |
|---|---|

- 13 Compared to the boiling point and the freezing point of water at 1 atmosphere, a 1.0 M $\text{CaCl}_2(\text{aq})$ solution at 1 atmosphere has a
- lower boiling point and a lower freezing point
 - lower boiling point and a higher freezing point
 - higher boiling point and a lower freezing point
 - higher boiling point and higher freezing point
- 14 According to the kinetic molecular theory, which statement describes an ideal gas?
- The gas particles are diatomic.
 - Energy is created when the gas particles collide.
 - There are no attractive forces between the gas particles.
 - The distance between the gas particles is small, compared to their size.
- 15 Which physical change is endothermic?
- $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$
 - $\text{CO}_2(\text{l}) \rightarrow \text{CO}_2(\text{s})$
 - $\text{CO}_2(\text{g}) \rightarrow \text{CO}_2(\text{l})$
 - $\text{CO}_2(\text{g}) \rightarrow \text{CO}_2(\text{s})$
- 16 Which Group 16 element combines with hydrogen to form a compound that has the strongest hydrogen bonding between its molecules?
- oxygen
 - selenium
 - sulfur
 - tellurium
- 17 Hydrocarbons are composed of the elements
- carbon and hydrogen, only
 - carbon and oxygen, only
 - carbon, hydrogen, and oxygen
 - carbon, nitrogen, and oxygen
- 18 Which atom is bonded to the carbon atom in the functional group of a ketone?
- fluorine
 - hydrogen
 - nitrogen
 - oxygen
- 19 Two types of organic reactions are
- addition and sublimation
 - deposition and saponification
 - decomposition and evaporation
 - esterification and polymerization
- 20 The isomers butane and methylpropane have
- the same molecular formula and the same properties
 - the same molecular formula and different properties
 - different molecular formulas and the same properties
 - different molecular formulas and different properties
- 21 In a redox reaction, which particles are lost and gained in equal numbers?
- electrons
 - neutrons
 - hydroxide ions
 - hydronium ions
- 22 What is the oxidation state for a Mn atom?
- 0
 - +7
 - +3
 - +4
- 23 Which compounds are classified as electrolytes?
- KNO_3 and H_2SO_4
 - KNO_3 and CH_3OH
 - CH_3OCH_3 and H_2SO_4
 - CH_3OCH_3 and CH_3OH
- 24 Which compound is an Arrhenius base?
- CO_2
 - CaSO_4
 - $\text{Ca}(\text{OH})_2$
 - $\text{C}_2\text{H}_5\text{OH}$
- 25 According to one acid-base theory, a water molecule acts as a base when it accepts
- an H^+ ion
 - an OH^- ion
 - a neutron
 - an electron

- 26 Given the equation representing a system at equilibrium:



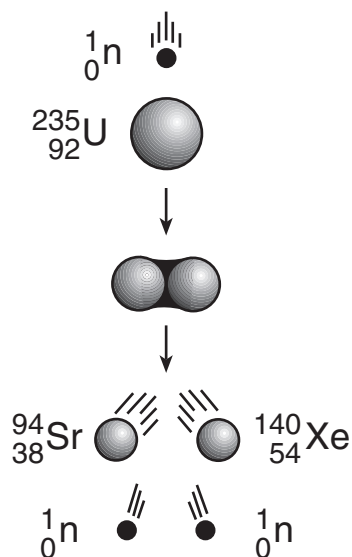
Which statement describes this reaction at equilibrium?

- (1) The concentration of $\text{N}_2(\text{g})$ decreases.
 - (2) The concentration of $\text{N}_2(\text{g})$ is constant.
 - (3) The rate of the reverse reaction decreases.
 - (4) The rate of the reverse reaction increases.
- 27 The acidity or alkalinity of an unknown aqueous solution is indicated by its
- (1) pH value
 - (2) electronegativity value
 - (3) percent by mass concentration
 - (4) percent by volume concentration
- 28 The laboratory process in which the volume of a solution of known concentration is used to determine the concentration of another solution is called
- (1) distillation
 - (2) fermentation
 - (3) titration
 - (4) transmutation

- 29 Which list of nuclear emissions is arranged in order from the greatest penetrating power to the least penetrating power?

- (1) alpha particle, beta particle, gamma ray
- (2) alpha particle, gamma ray, beta particle
- (3) gamma ray, alpha particle, beta particle
- (4) gamma ray, beta particle, alpha particle

- 30 Given the diagram representing a reaction:



Which type of change is represented?

- (1) fission
- (2) fusion
- (3) deposition
- (4) evaporation

Part B-1

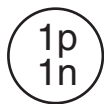
Answer all questions in this part.

Directions (31–50): For each statement or question, record on your separate answer sheet the number of the word or expression that, of those given, best completes the statement or answers the question. Some questions may require the use of the 2011 Edition Reference Tables for Physical Setting/Chemistry.

- 31 Which electron shell contains the valence electrons of a radium atom in the ground state?

(1) the sixth shell (3) the seventh shell
(2) the second shell (4) the eighteenth shell

- 32 Each diagram below represents the nucleus of an atom.



How many different elements are represented by the diagrams?

(1) 1 (3) 3
(2) 2 (4) 4

- 33 Chlorine and element X have similar chemical properties. An atom of element X could have an electron configuration of

(1) 2-2 (3) 2-8-8
(2) 2-8-1 (4) 2-8-18-7

- 34 Which group of elements contains a metalloid?

(1) Group 8 (3) Group 16
(2) Group 2 (4) Group 18

- 35 Which Lewis electron-dot diagram represents a fluoride ion?



(1)



(2)



(3)



(4)

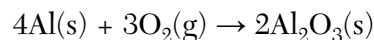
- 36 In the formula for the compound XCl_4 , the X could represent

(1) C (3) Mg
(2) H (4) Zn

- 37 The formula C_2H_4 can be classified as

(1) a structural formula, only
(2) a molecular formula, only
(3) both a structural formula and an empirical formula
(4) both a molecular formula and an empirical formula

- 38 Given the balanced equation representing a reaction:



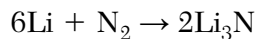
How many moles of Al(s) react completely with 4.50 moles of $\text{O}_2\text{(g)}$ to produce 3.00 moles of $\text{Al}_2\text{O}_3\text{(s)}$?

(1) 1.50 mol (3) 6.00 mol
(2) 2.00 mol (4) 4.00 mol

- 39 What is the percent composition by mass of oxygen in $\text{Ca(NO}_3)_2$ (gram-formula mass = 164 g/mol)?

(1) 9.8% (3) 48%
(2) 29% (4) 59%

- 40 Given the balanced equation representing a reaction:



Which type of chemical reaction is represented by this equation?

- (1) synthesis (3) single replacement
(2) decomposition (4) double replacement
- 41 Which elements can react to produce a molecular compound?
- (1) calcium and chlorine
(2) hydrogen and sulfur
(3) lithium and fluorine
(4) magnesium and oxygen
- 42 Compared to a 1.0-mole sample of NaCl(s) , a 1.0-mole sample of NaCl(l) has a *different*
- (1) number of ions
(2) empirical formula
(3) gram-formula mass
(4) electrical conductivity
- 43 Which property of an unsaturated solution of sodium chloride in water remains the same when more water is added to the solution?
- (1) density of the solution
(2) boiling point of the solution
(3) mass of sodium chloride in the solution
(4) percent by mass of water in the solution
- 44 Which ion combines with Ba^{2+} to form a compound that is most soluble in water?
- (1) S^{2-} (3) CO_3^{2-}
(2) OH^- (4) SO_4^{2-}

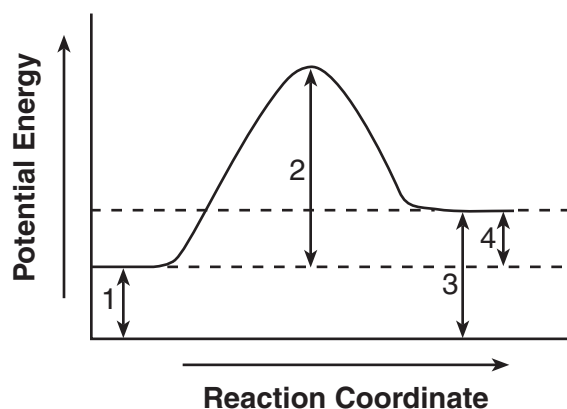
- 45 When a sample of gas is cooled in a sealed, rigid container, the pressure the gas exerts on the walls of the container will decrease because the gas particles hit the walls of the container

(1) less often and with less force
(2) less often and with more force
(3) more often and with less force
(4) more often and with more force

- 46 A rigid cylinder with a movable piston contains 50.0 liters of a gas at 30.0°C with a pressure of 1.00 atmosphere. What is the volume of the gas in the cylinder at STP?

(1) 5.49 L (3) 55.5 L
(2) 45.0 L (4) 455 L

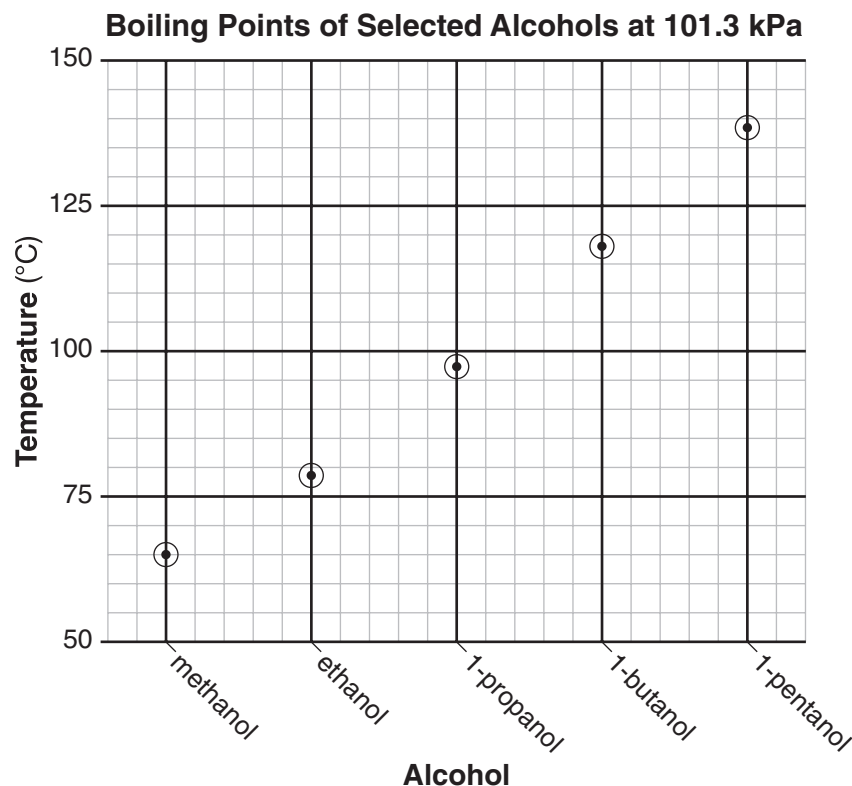
- 47 Given the potential energy diagram for a chemical reaction:



Which numbered interval represents the heat of reaction?

(1) 1 (3) 3
(2) 2 (4) 4

Base your answers to questions 48 and 49 on the graph below and on your knowledge of chemistry.



48 What is represented by the number “1” in the IUPAC name for three of these alcohols?

- (1) the number of isomers for each alcohol
- (2) the number of -OH groups for each carbon atom in each alcohol molecule
- (3) the location of an -OH group on one end of the carbon chain in each alcohol molecule
- (4) the location of an -OH group in the middle of the carbon chain in each alcohol molecule

49 What can be concluded from this graph?

- (1) At 101.3 kPa, water has a higher boiling point than 1-butanol.
 - (2) At 101.3 kPa, water has a lower boiling point than ethanol.
 - (3) The greater the number of carbon atoms per alcohol molecule, the lower the boiling point of the alcohol.
 - (4) The greater the number of carbon atoms per alcohol molecule, the higher the boiling point of the alcohol.
-

- 50 In the laboratory, a student investigates the effect of concentration on the reaction between HCl(aq) and Mg(s) , changing only the concentration of HCl(aq) . Data for two trials in the investigation are shown in the table below.

Data Table

Trial	Volume of HCl(aq) (mL)	Concentration of HCl(aq) (M)	Mass of Mg(s) (g)	Reaction Time (s)
1	50.0	0.2	0.1	48
2	50.0	0.4	0.1	?

Compared to trial 1, what is the expected reaction time for trial 2 and the explanation for that result?

- (1) less than 48 s, because there are fewer effective particle collisions per second
 - (2) less than 48 s, because there are more effective particle collisions per second
 - (3) more than 48 s, because there are fewer effective particle collisions per second
 - (4) more than 48 s, because there are more effective particle collisions per second
-

Part B–2

Answer all questions in this part.

Directions (51–65): Record your answers in the spaces provided in your answer booklet. Some questions may require the use of the *2011 Edition Reference Tables for Physical Setting/Chemistry*.

- 51 Determine the volume of 2.00 M HCl(aq) solution required to completely neutralize 20.0 milliliters of 1.00 M NaOH(aq) solution. [1]
- 52 Determine the mass of KNO₃ that dissolves in 100. grams of water at 40.°C to produce a saturated solution. [1]
- 53 State, in terms of molecular polarity, why ethanol is soluble in water. [1]

Base your answers to questions 54 through 56 on the information below and on your knowledge of chemistry.

Three elements, represented by *D*, *E*, and *Q*, are located in Period 3. Some properties of these elements are listed in the table below. A student's experimental result indicates that the density of element *Q* is 2.10 g/cm³, at room temperature and standard pressure.

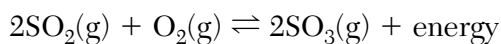
**Properties of Samples of Three Elements
at Room Temperature and Standard Pressure**

Element	Phase	Mass (g)	Density (g/cm ³)	Oxide Formula
D	solid	50.0	0.97	D ₂ O
E	solid	50.0	1.74	EO
Q	solid	50.0	2.00	QO ₂ or QO ₃

- 54 Identify the physical property in the table that could be used to differentiate the samples of the three elements from each other. [1]
- 55 Identify the group on the Periodic Table to which element *D* belongs. [1]
- 56 Determine the percent error between the student's experimental density and the accepted density of element *Q*. [1]
-

Base your answers to questions 57 through 59 on the information below and on your knowledge of chemistry.

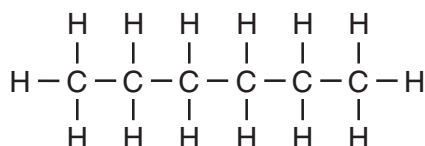
The equation below represents an equilibrium system of $\text{SO}_2(\text{g})$, $\text{O}_2(\text{g})$, and $\text{SO}_3(\text{g})$. The reaction can be catalyzed by vanadium or platinum.



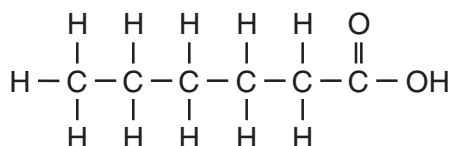
- 57 Compare the rates of the forward and reverse reactions at equilibrium. [1]
- 58 State how the equilibrium shifts when $\text{SO}_3(\text{g})$ is removed from the system. [1]
- 59 A potential energy diagram for the forward reaction is shown *in your answer booklet*. On this diagram, draw a dashed line to show how the potential energy changes when the reaction occurs by the catalyzed pathway. [1]
-

Base your answers to questions 60 and 61 on the information below and on your knowledge of chemistry.

The formulas for two compounds are shown below.



Compound A



Compound B

- 60 Explain, in terms of bonding, why compound A is saturated. [1]
- 61 Explain, in terms of molecular structure, why the chemical properties of compound A are different from the chemical properties of compound B. [1]
-

Base your answers to questions 62 through 65 on the information below and on your knowledge of chemistry.

Some isotopes of potassium are K-37, K-39, K-40, K-41, and K-42. The natural abundance and the atomic mass for the naturally occurring isotopes of potassium are shown in the table below.

Naturally Occurring Isotopes of Potassium

Isotope Notation	Natural Abundance (%)	Atomic Mass (u)
K-39	93.26	38.96
K-40	0.01	39.96
K-41	6.73	40.96

- 62 Identify the decay mode of K-37. [1]
- 63 Complete the nuclear equation *in your answer booklet* for the decay of K-40 by writing a notation for the missing nuclide. [1]
- 64 Determine the fraction of an original sample of K-42 that remains unchanged after 24.72 hours. [1]
- 65 Show a numerical setup for calculating the atomic mass of potassium. [1]
-

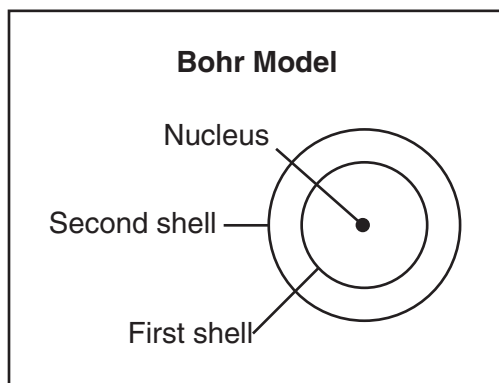
Part C

Answer all questions in this part.

Directions (66–85): Record your answers in the spaces provided in your answer booklet. Some questions may require the use of the *2011 Edition Reference Tables for Physical Setting/Chemistry*.

Base your answers to questions 66 through 68 on the information below and on your knowledge of chemistry.

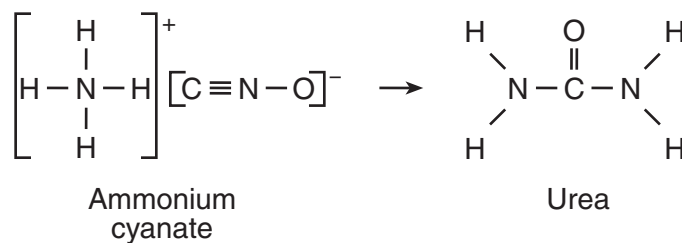
The Bohr model of the atom was developed in the early part of the twentieth century. A diagram of the Bohr model for one atom, in the ground state, of a specific element is shown below. The nucleus of this atom contains 4 protons and 5 neutrons.



- 66 State the atomic number and the mass number of this element. [1]
- 67 State the number of electrons in *each* shell in this atom in the ground state. [1]
- 68 Using the Bohr model, describe the changes in electron energy and electron location when an atom changes from the ground state to an excited state. [1]
-

Base your answers to questions 69 through 72 on the information below and on your knowledge of chemistry.

In 1828, Friedrich Wöhler produced urea when he heated a solution of ammonium cyanate. This reaction is represented by the balanced equation below.



69 Identify the element in urea that makes it an organic compound. [1]

70 Determine the gram-formula mass of the product. [1]

71 Write an empirical formula for the product. [1]

72 Explain why this balanced equation represents a conservation of atoms. [1]

Base your answers to questions 73 through 75 on the information below and on your knowledge of chemistry.

Rubbing alcohol sold in stores is aqueous 2-propanol, $\text{CH}_3\text{CHOHCH}_3(\text{aq})$. Rubbing alcohol is available in concentrations of 70.% and 91% 2-propanol by volume.

To make 100. mL of 70.% aqueous 2-propanol, 70. mL of 2-propanol is diluted with enough water to produce a total volume of 100. mL. In a laboratory investigation, a student is given a 132-mL sample of 91% aqueous 2-propanol to separate using the process of distillation.

73 State evidence that indicates the proportions of the components in rubbing alcohol can vary. [1]

74 Identify the property of the components that makes it possible to use distillation to separate the 2-propanol from water. [1]

75 Determine the maximum volume of 2-propanol in the 132-mL sample. [1]

Base your answers to questions 76 through 79 on the information below and on your knowledge of chemistry.

A sample of seawater is analyzed. The table below gives the concentration of some ions in the sample.

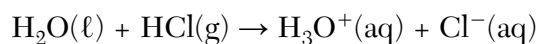
**Concentration of Some Ions
in a Seawater Sample**

Ion	Concentration (M)
Cl ⁻	0.545
Na ⁺	0.468
Mg ²⁺	0.054
SO ₄ ²⁻	0.028
Ca ²⁺	0.010
K ⁺	0.010

- 76 Write a chemical formula of *one* compound formed by the combination of K⁺ ions with one of these ions as water completely evaporates from the seawater sample. [1]
- 77 Determine the number of moles of the SO₄²⁻ ion in a 1400.-liter sample of the seawater. [1]
- 78 Compare the radius of an Mg²⁺ ion in the seawater to the radius of an Mg atom. [1]
- 79 Using the key *in your answer booklet*, draw *two* water molecules in the box, showing the orientation of *each* water molecule toward the calcium ion. [1]
-

Base your answers to questions 80 through 82 on the information below and on your knowledge of chemistry.

A scientist bubbled HCl(g) through a sample of H₂O(l). This process is represented by the balanced equation below.

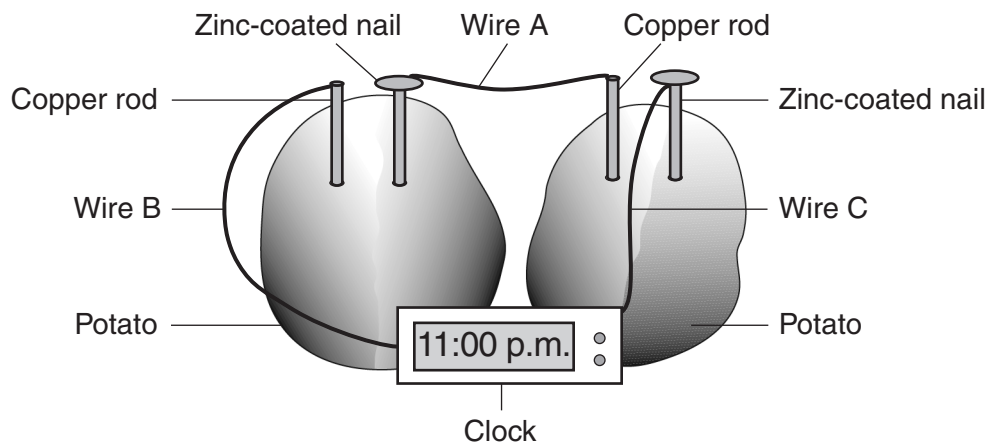


The scientist measured the pH of the liquid in the flask before and after the gas was bubbled through the water. The initial pH value of the water was 7.0 and the final pH value of the solution was 3.0.

- 80 Explain, in terms of ions, why the gaseous reactant in the equation is classified as an Arrhenius acid. [1]
- 81 What would be the color of bromcresol green if it had been added to the water in the flask before any of the HCl(g) was bubbled through the water? [1]
- 82 Compare the hydronium ion concentration of the solution that has the pH value of 3.0 to the hydronium ion concentration of the water. [1]
-

Base your answers to questions 83 through 85 on the information below and on your knowledge of chemistry.

A small digital clock can be powered by a battery made from two potatoes and some household materials. The “potato clock” battery consists of two cells connected in a way to produce enough electricity to allow the clock to operate. In each cell, zinc atoms react to form zinc ions. Hydrogen ions from phosphoric acid in the potatoes react to form hydrogen gas. The labeled diagram and balanced ionic equation below show the reaction, the materials, and connections necessary to make a “potato clock” battery.



83 State the direction of electron flow in wire A as the two cells operate. [1]

84 Write a balanced half-reaction equation for the oxidation that occurs in the “potato clock” battery. [1]

85 Explain why phosphoric acid is needed for the battery to operate. [1]

FOR TEACHERS ONLY

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

P.S.–CH PHYSICAL SETTING/CHEMISTRY

Tuesday, June 23, 2015 — 9:15 a.m. to 12:15 p.m., only

SCORING KEY AND RATING GUIDE

Directions to the Teacher:

Refer to the directions on page 2 before rating student papers.

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

Part A and Part B–1

Allow 1 credit for each correct response.

Part A			
1 4	9 4	17 1	25 1
2 1	10 2	18 4	26 2
3 4	11 2	19 4	27 1
4 4	12 4	20 2	28 3
5 2	13 3	21 1	29 4
6 4	14 3	22 1	30 1
7 1	15 1	23 1	
8 1	16 1	24 3	
Part B–1			
31 3	36 1	41 2	46 2
32 2	37 2	42 4	47 4
33 4	38 3	43 3	48 3
34 3	39 4	44 2	49 4
35 1	40 1	45 1	50 2

Map to Core Curriculum

June 2015 Physical Setting/Chemistry			
Question Numbers			
Key Ideas/Performance Indicators	Part A	Part B	Part C
Standard 1			
Math Key Idea 1		39, 46, 51, 52, 56, 59, 65	70
Math Key Idea 2		47, 49	71
Math Key Idea 3		38, 39, 46, 51, 63, 64	70, 75, 76, 77
Science Inquiry Key Idea 1		42, 45, 48, 50, 53, 54, 55, 61	69, 72, 73, 74, 78, 79, 80, 83, 85
Science Inquiry Key Idea 2			
Science Inquiry Key Idea 3		31, 32, 37, 41, 43, 44, 48, 49, 50, 54, 62	66, 67, 68, 72, 76, 81, 84
Engineering Design Key Idea 1			
Standard 2			
Key Idea 1			
Key Idea 2			73
Key Idea 3			
Standard 6			
Key Idea 1			
Key Idea 2		32, 45, 56, 59, 60, 61	79
Key Idea 3			82
Key Idea 4		58	
Key Idea 5		52	
Standard 7			
Key Idea 1			
Key Idea 2			
Standard 4 Process Skills			
Key Idea 3		31, 33, 34, 36, 37, 38, 40, 45, 46, 50, 51, 52, 54, 55, 57, 58, 65	66, 67, 68, 70, 71, 72, 74, 75, 77, 81, 83, 84
Key Idea 4		47, 59, 63, 64	
Key Idea 5		35, 49, 53	79
Standard 4			
Key Idea 3	1, 2, 3, 4, 10, 11, 12, 13, 14, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29	31, 32, 33, 34, 36, 37, 38, 39, 40, 42, 43, 44, 45, 46, 48, 50, 51, 52, 54, 55, 56, 57, 58, 59, 60, 61, 65	66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 80, 81, 82, 83, 84, 85
Key Idea 4	15, 30	47, 62, 63, 64	
Key Idea 5	5, 6, 7, 8, 9, 16	35, 41, 49, 53	78, 79
Reference Tables			
2011 Edition	1, 3, 6, 7, 8, 11, 12, 16, 17, 18, 24, 27, 29	31, 32, 33, 34, 35, 36, 39, 41, 44, 46, 48, 49, 51, 52, 55, 56, 60, 61, 62, 63, 64	66, 67, 68, 70, 75, 76, 77, 78, 79, 81, 83, 84

Part B–2

Allow a total of 15 credits for this part. The student must answer all questions in this part.

51 [1] Allow 1 credit for 10.0 mL. Significant figures do *not* need to be shown.

52 [1] Allow 1 credit for 64 g, or any value from 62 g to 66 g, inclusive.

53 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Both ethanol molecules and water molecules are polar.

Water molecules and ethanol molecules have similar polarity.

54 [1] Allow 1 credit for density.

55 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Group 1

alkali metals

56 [1] Allow 1 credit for 5.0%. Significant figures do *not* need to be shown.

57 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

At equilibrium, the rates of the forward and reverse reactions are equal.

The rates are the same.

58 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The equilibrium will shift to favor the formation of SO_3 .

The rate of the forward reaction is greater than the rate of the reverse reaction.

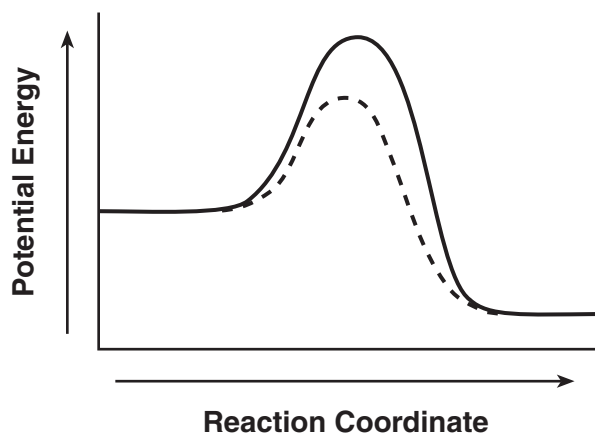
The equilibrium will shift to favor the forward reaction.

The equilibrium will shift to the right.

The concentrations of the reactants will decrease.

59 [1] Allow 1 credit.

Example of a 1-credit response:



60 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

All the carbon-to-carbon bonds are single bonds.

The maximum number of H atoms are bonded to the carbon chain.

There are no multiple bonds.

61 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

A molecule of compound *B* has an organic acid functional group and a molecule of compound *A* has no functional group.

A molecule of *A* has only single bonds and a molecule of *B* has one double-bonded oxygen atom.

A molecule of compound *B* has two O atoms and a molecule of compound *A* has no O atoms in its structure.

A is a hydrocarbon but *B* is an acid.

A is an alkane but *B* is an acid.

62 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

positron decay

β^+

${}_{+1}^0\text{e}$

63 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

${}_{20}^{40}\text{Ca}$

Ca-40

calcium-40

64 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

$\frac{1}{4}$

25%

0.25

65 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

$$\frac{(93.26)(38.96 \text{ u}) + (0.01)(39.96 \text{ u}) + (6.73)(40.96 \text{ u})}{100}$$

$$(93.26\%)(38.96 \text{ u}) + (0.01\%)(39.96 \text{ u}) + (6.73\%)(40.96 \text{ u})$$

$$(0.9326)(38.96) + (0.0673)(40.96) + (0.0001)(39.96)$$

Part C

Allow a total of 20 credits for this part. The student must answer all questions in this part.

66 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Atomic number: 4

Mass number: 9

67 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Number of electrons in first shell: 2 or $2e^{-}$

Number of electrons in second shell: 2 or $2e^{-}$

68 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Change in electron energy:

Electron energy increases.

An electron absorbs energy.

more energy

Change in electron location:

An electron moves to a higher electron shell.

from the first to the second shell

second to higher energy level

farther from the nucleus

69 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

carbon

C

70 [1] Allow 1 credit for 60. g/mol. Significant figures do *not* need to be shown.

71 [1] Allow 1 credit for $\text{CH}_4\text{N}_2\text{O}$. The order of the elements can vary.

72 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

There are the same number of atoms of each element on both sides of the equation.

No atoms are lost or gained.

73 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The aqueous solutions of 2-propanol do not contain the same proportions of alcohol and water.
Rubbing alcohol is sold as 70.% and 91% solutions.

74 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The water and the 2-propanol have different boiling points.
strength of the intermolecular forces
boiling point
vapor pressure

75 [1] Allow 1 credit for 120 mL. Significant figures do *not* need to be shown.

76 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

KCl
K₂SO₄

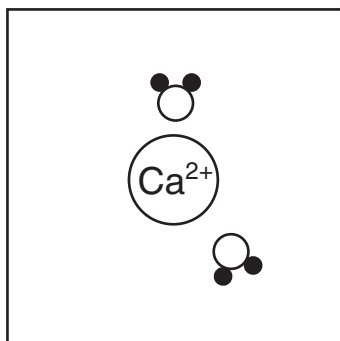
77 [1] Allow 1 credit for 39 mol. Significant figures do *not* need to be shown.

78 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The radius of an Mg²⁺ ion is smaller than the radius of an Mg atom.
The atom has a larger radius than the ion.

79 [1] Allow 1 credit. Acceptable responses must show *at least two* water molecules with the oxygen atom of *each* water molecule oriented toward the calcium ion.

Example of a 1-credit response:



80 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The HCl(g) is an Arrhenius acid because it yields $\text{H}_3\text{O}^+(\text{aq})$ ions.

The gaseous reactant increases the concentration of $\text{H}^+(\text{aq})$.

It produces hydrogen ions in water.

81 [1] Allow 1 credit for blue.

82 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The hydronium ion concentration of the acidic solution is 10^4 times greater than the hydronium ion concentration in the water.

The H^+ concentration is less in the water.

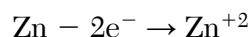
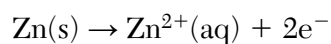
83 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

from the zinc-coated nail to the copper rod

from Zn to Cu

left to right

84 [1] Allow 1 credit. Acceptable responses include, but are not limited to:



85 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The H^+ ions that are reduced come from the phosphoric acid.

Phosphoric acid releases ions that are free to move.

H_3PO_4 is an electrolyte.

$\text{H}_3\text{PO}_4(\text{aq})$ can act as a salt bridge.